

STUDENT GRADE CALCULATOR

Python Programming Project Statement

Project Type	First-Year Python Project
Programming Language	Python
Application Type	Menu-Driven Student Record and Grade Calculation Simulation
Development Platform	Google Colab / Python IDE

Project Summary

The Student Grade Calculator is a beginner-friendly Python application developed to simulate a small result-processing desk of a school or college. The program allows users to add a student with the marks of five subjects, calculate the percentage and grade automatically, display all student records, search a student by roll number and show an individual result through a simple menu-driven interface.

The project focuses on applying fundamental Python programming concepts to a real-life problem. It does not connect to a real examination system, database or university portal; it is intended only for academic learning and demonstration.

Submitted as a Python Programming Project

1. Introduction

Result processing is a common example of a computer-based information system. Schools and colleges use it to record marks, calculate percentages, assign grades and let students or teachers look up results. This project creates a simplified version of such a system using Python.

The program is designed around a menu. The user selects an operation, provides the required information, and receives an appropriate result. After completing an operation, the menu appears again until the user chooses to exit.

2. Problem Statement

Design and implement a Python program that can accept the name, roll number and marks of five subjects for a student, calculate the percentage, assign a grade according to fixed slabs, and store the record. The system should also be able to display all records, search a student by roll number and show the result of a single student.

3. Objectives

- To develop a simple real-life application using Python.
- To understand how dictionaries can store structured student information.
- To use lists to hold the marks of a student and the collection of all students.
- To use functions to divide the program into manageable modules.
- To apply conditional statements for decision making, such as grade slabs.
- To use loops for repeated menu operations and for reading marks.
- To perform arithmetic operations for total, average and percentage.

4. Scope of the Project

The current project is limited to a single session, five subjects per student and in-memory data storage. It is suitable for academic demonstration and learning. It can later be extended with file handling, databases, editing and deleting records, subject names, rank lists and a graphical user interface.

5. System Features and Working

5.1 Add Student

The user enters the student name, the roll number and the marks of five subjects. The program calculates the percentage (average of the five marks), decides the grade, stores everything as one record in the students list and confirms with a success message.

5.2 Display All Students

The program shows the name, roll number, marks, percentage and grade of every stored student. If no student has been added, it reports that no records were found.

5.3 Search Student

The user enters a roll number. If a matching record exists, its full details are displayed; otherwise the program reports that the student was not found.

5.4 Show Result

The user enters a roll number and the program prints a short result card containing the name, roll number, percentage and grade. An unknown roll number gives a not-found message.

5.5 Grade Slabs

Percentage	Grade
90 or more	A+
80 to below 90	A
70 to below 80	B
60 to below 70	C
50 to below 60	D
Below 50	F

5.6 Exit

The program ends with a thank-you message.

6. Program Flow

Step	Operation
1	Start the program
2	Display the main menu
3	Accept the user's choice
4	Perform the selected operation
5	Search the students list where required
6	Display the result
7	Return to the main menu
8	Exit when the user selects Exit

Main Menu

```
=====
STUDENT GRADE MANAGEMENT
=====
1. Add Student
2. Display All Students
3. Search Student
4. Show Result
5. Exit
=====
```

7. Python Concepts Used

Functions

Separate functions are used for calculating the grade, adding a student, displaying all students, searching a student and showing a result. This makes the program easier to read and maintain.

Dictionary and List

Each student is stored as a dictionary with the keys name, roll, marks, percentage and grade. A list named 'students' holds all the student dictionaries, and a list named 'marks' holds the five subject marks of one student.

Conditional Statements

if, elif and else statements are used to select menu options and to choose the grade from the percentage.

Loops

A while loop keeps the menu running until the user chooses the Exit option. A for loop reads the five marks and another for loop walks through the students list for display and search.

Input and Arithmetic Operations

The input() function accepts information from the user, float() converts marks to numbers, and sum(), len() and division are used to find the percentage.

8. Sample Test Cases

Test	Input	Expected Result
Add student	Aman Raj, roll 101, marks 95, 92, 88, 90, 95	Student added successfully!
Percentage and grade	Display all students	Percentage 92.0 %, Grade A+
Grade A	Marks 80, 80, 80, 80, 80	Percentage 80.0 %, Grade A
Grade B	Marks 70 in all subjects	Percentage 70.0 %, Grade B
Grade C	Marks 60 in all subjects	Percentage 60.0 %, Grade C
Grade D	Marks 50 in all subjects	Percentage 50.0 %, Grade D
Grade F	Marks 40, 45, 30, 50, 35	Percentage 40.0 %, Grade F
Search student	Roll number 101	Student Found! with full details
Unknown roll number	Roll number 999	Student not found.
Empty records	Option 2 with no students	No student records found.

9. Advantages

- Simple and beginner-friendly interface.
- Uses important Python programming concepts.

- Easy to modify and expand.
- Calculates percentage and grade automatically, avoiding manual errors.
- Provides clear messages for common mistakes.

10. Limitations

- Data is stored only while the program is running.
- Non-numeric input for marks is not handled and stops the program.
- Marks are not checked to be between 0 and 100.
- Every student must have exactly five subjects, and subjects have no names.
- Records cannot be edited or deleted.
- The project is intended for educational use only.

11. Future Enhancements

- Add try/except handling and marks range validation (0 to 100).
- Reject duplicate roll numbers.
- Add options to edit and delete a student record.
- Save records permanently in a file or database.
- Add subject names, rank list and class average.
- Develop a graphical user interface using Tkinter.

12. Learning Outcomes

This project provides practical experience in converting a real-life requirement into a structured Python program. It improves understanding of functions, dictionaries, lists, loops, conditional statements, input handling and basic problem solving.

13. Conclusion

The Student Grade Calculator demonstrates how fundamental Python programming concepts can be combined to create a useful menu-driven application. The system performs the core steps of a result desk: recording marks, calculating the percentage, assigning a grade, and displaying or searching student results.

Although the project is a simplified simulation and does not provide the validation, persistence or security features of a real examination system, it provides a strong foundation for future development.

Project Status: Completed – Educational Python Simulation

End of Project Statement